

Chapter 2: Electric Fields, Potential & DC Circuits

2.1 Coulomb's Law and Electrostatic Force

Inverse-Square Law for Point Charges

Electrostatics investigates electric charges at rest. Charles-Augustin de Coulomb demonstrated through torsion balance experiments that the electrostatic interaction force between two stationary point charges is directly proportional to the product of the magnitudes of the charges and inversely proportional to the square of the separation distance between them.

Key Formula / Law: Coulomb's Law: $F = k \cdot (|q_1 \cdot q_2|) / r^2$ where $k = 1 / (4\pi\epsilon_0) \approx 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$

2.2 Electric Field & Electric Potential

Vector Field and Scalar Energy Landscape

An electric charge alters the space surrounding it, establishing an electric field $E = F / q_{\text{test}}$ (in N/C or V/m). The work required per unit test charge to move from infinity to a point in space defines the Electric Potential V .

Capacitance measures the capacity of a conductor system to store electrostatic charge per unit potential difference: $C = Q / V$. For a parallel-plate capacitor with plate area A and separation d , $C = \epsilon_0 A / d$.

2.3 Direct Current (DC) Circuits & Kirchhoff's Laws

Current, Resistance, and Conservation Principles

Electric current I is the net rate of flow of electric charge: $I = dQ/dt$ (Amperes). Ohm's law specifies that for ohmic conductors, $V = I \cdot R$.

For complex multi-loop circuits, Gustav Kirchhoff formulated two conservation rules: 1. Kirchhoff's Current Law (Junction Rule): The algebraic sum of all currents entering a junction equals the sum of currents leaving (Conservation of Charge, $\Sigma I = 0$). 2. Kirchhoff's Voltage Law (Loop Rule): The algebraic sum of changes in potential around any closed circuit loop is zero (Conservation of Energy, $\Sigma \Delta V = 0$).

Key Takeaways:

- Electric field points away from positive source charges and toward negative charges.
- Electrostatic force is conservative; hence line integral of E around any closed loop is zero.
- Capacitors store potential energy $U = 1/2 C V^2$.
- Kirchhoff's Junction rule reflects conservation of charge; Loop rule reflects conservation of energy.